Project Initialization and Planning Phase

Date	28 September 2024
Team ID	PNT2024TMID587921
Project Title	Streamlining IT Procurement: Automating standard laptop orders with WI
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal report presents a practical solution to streamline the IT procurement process for standard laptop orders. The system uses an automated workflow (WI) that connects procurement requests, manager approvals, budget verification, vendor selection and inventory updates. The goal is to reduce delays, cut manual errors and make the whole ordering cycle faster and more transparent for both employees and the IT procurement team.

1. Abstract

IT departments in most organisations still rely on manual or semi-manual methods to order standard laptops for employees. This leads to long approval chains, missing budget checks, duplicate orders and poor inventory visibility. The proposed project builds an automated workflow system that handles the complete life-cycle of a standard laptop order – from raising a procurement request to final inventory update. Key steps such as manager approval, budget check, preferred-vendor selection and stock allocation are performed automatically with clear status tracking. The solution aims to improve operational efficiency, reduce processing time and maintain better control over IT asset spending.

2. Introduction

Every organisation needs laptops for new joiners, replacements and role changes. In a typical IT setup these requests travel through emails, paper forms or simple ticketing tools. Managers approve the request, finance verifies the budget, the procurement team contacts vendors and finally the asset is recorded in inventory. Because each step is handled by different people using different tools, the process becomes slow and error-prone.

An automated workflow integrates all these steps into one controlled path. When an employee raises a request for a standard laptop configuration, the system routes it to the right approver, checks the available budget, looks at current stock, selects the approved vendor and creates the purchase order – all with minimal human intervention. This project focuses on designing and describing such a workflow for standard laptop orders.

3. Problem Statement

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Description	The current laptop ordering process is largely manual. Employees send requests by email or fill shared spreadsheets. Manager approval is often delayed because of busy inboxes. Budget checks are done late or missed, leading to overspending. Vendor selection is inconsistent and inventory records are updated only after delivery, causing stock mismatches.
Impact	These issues result in longer waiting times for employees, higher administrative effort for the procurement team, occasional budget overruns and inaccurate asset registers. Overall, the organisation loses time and money while employee satisfaction also suffers.

4. Objectives

The main objectives of this project are:

- To design an automated workflow that covers the complete standard laptop order cycle.
- To enable employees to raise a procurement request through a simple digital form.
- To route the request automatically to the reporting manager for approval.
- To perform an automatic budget check against the department's remaining allocation.
- To select the preferred vendor based on price agreements and current inventory status.
- To update the inventory system once the order is confirmed or the laptop is issued.
- To provide real-time status visibility to the requester and the procurement team.

5. Existing System

In the existing system the laptop ordering process depends heavily on human coordination:

- Employee sends a request email or fills a shared Excel sheet with laptop specifications.
- Manager receives the mail and replies with approval or rejection – often after several reminders.
- Procurement team manually checks whether the department still has budget.
- If budget is available, the team contacts one or more vendors by phone or email for quotations.
- After vendor selection a purchase order is prepared offline and sent to the vendor.
- When the laptop arrives, inventory is updated in a separate spreadsheet or asset tool – sometimes days later.
- There is no single place where the requester can see the current status of the order.

Because the steps are disconnected, errors such as double ordering, wrong configuration and missed budget limits occur regularly. The process is also slow – a simple standard laptop order can take two to three weeks.

6. Proposed System

The proposed system introduces a single automated workflow (WI) that connects all the required steps for ordering a standard laptop. The system is designed around clear roles and decision points so that each action happens only when the previous condition is satisfied.

Core features of the proposed system:

- Digital procurement request form with fixed standard laptop models and configurations.
- Automatic routing of the request to the employee's reporting manager for approval.
- Real-time budget check that compares the laptop cost against the department's remaining budget.
- Automatic inventory look-up – if a matching laptop is already in stock it can be issued directly.
- Rule-based vendor selection using pre-approved vendor list and negotiated prices.
- Generation of purchase order and notification to the selected vendor.
- Automatic update of inventory and asset records after order confirmation or delivery.
- Status dashboard for the requester, manager and procurement team.

7. System Architecture & Workflow

The system follows a linear yet flexible workflow. Each step is triggered only after the previous step is successfully completed. Below is the step-by-step laptop order automation:

Step 1 – Procurement Request

An employee logs into the internal portal and selects "Request Standard Laptop". He/she chooses the approved model (for example, basic, intermediate or advanced configuration), enters the business justification and submits the request. The system creates a unique request ID.

Step 2 – Manager Approval

The request is automatically assigned to the employee's reporting manager. The manager receives a notification and can approve or reject the request with comments. If rejected, the requester is informed and the workflow ends. If approved, the request moves to the next stage.

Step 3 – Budget Check

Once approved, the system queries the finance or cost-centre database to check whether the department still has enough budget for the selected laptop cost. If the budget is insufficient, the request is returned to the manager or escalated to finance for additional approval. If the budget is available, the amount is reserved and the workflow continues.

Step 4 – Inventory Check

The system first looks into the IT inventory for an available laptop of the same standard configuration that is not assigned to any user. If such a unit exists, it is allocated to the employee, the inventory is updated and the request is closed as "Issued from Stock".

Step 5 – Vendor Selection

When no matching laptop is in stock, the system selects a vendor from the pre-approved vendor list. Selection rules can consider lowest negotiated price, delivery time and vendor rating. A purchase order is generated automatically with the laptop specification, quantity and delivery address.

Step 6 – Order Placement & Tracking

The purchase order is sent to the selected vendor by email or through the vendor portal. The procurement team and the requester can track the order status (confirmed, shipped, delivered). On delivery confirmation the laptop is received into inventory and then issued to the employee.

Step 7 – Inventory Update & Closure

Finally the system updates the asset register with the new laptop serial number, assigns it to the employee and closes the procurement request. All parties receive a completion notification. The reserved budget is converted into actual expenditure.

This end-to-end automated workflow reduces manual hand-offs, keeps every stakeholder informed and maintains a complete audit trail of the order.

8. Benefits

Faster processing: Standard laptop orders that earlier took weeks can be completed in a few days because approvals and checks happen automatically.

Reduced errors: Budget overruns, wrong configurations and missed inventory updates are minimised by system-enforced rules.

Better budget control: Real-time budget check and reservation prevent departments from exceeding their IT allocation.

Improved inventory accuracy: Stock is checked before ordering and updated immediately after issue or delivery.

Transparency: Requesters and managers can see the live status of every request without sending follow-up emails.

Lower administrative load: Procurement staff spend less time on routine standard orders and can focus on complex or exception cases.

Compliance & audit: Every step leaves a timestamped record that can be used for internal audits and policy compliance.

9. Conclusion

Manual handling of standard laptop orders creates delays, extra work and occasional financial leakage. By introducing an automated workflow that covers procurement request, manager approval, budget check, inventory verification, vendor selection and final asset update, the organisation can process these routine orders quickly and reliably. The proposed system keeps the process simple for employees while giving the IT

and finance teams better visibility and control. Overall it is a practical step toward digital and efficient IT procurement.

10. Future Scope

- Integrate the workflow with the organisation's HR system so that laptop requests are raised automatically for new joiners.
- Add predictive analytics to forecast laptop demand based on hiring plans and replacement cycles.
- Extend the same automation pattern to other standard IT items such as monitors, headsets and docking stations.
- Build a mobile-friendly interface so managers can approve requests on the go.
- Introduce AI-based anomaly detection to flag unusual request patterns or potential policy violations.
- Connect the system with major vendor APIs for real-time price and delivery-time updates.

11. References

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This document is prepared as part of the Project Initialization and Planning Phase. All content is original work for academic evaluation.